
Exact Gaussian Processes on the Sphere at Half a Million Points: Geodesic Non-Stationary Compactly-Supported Kernels for the Earth’s Gravity-Gradient Field

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Abstract

Gaussian process (GP) regression provides calibrated uncertainty quantification, but the $\mathcal{O}(N^2)$ storage and $\mathcal{O}(N^3)$ time of exact training restrict it to small datasets unless one accepts an approximation. Recent work has shown that the covariance matrix of a GP is not inherently dense: with a flexible, compactly-supported, non-stationary kernel the covariance becomes naturally sparse, and exact training reduces to sparse linear algebra. All kernels in that framework, however, are defined through a *Euclidean* distance on the input coordinates. Here we show that for global geophysical fields – fields defined on the sphere, with correlation lengths that change by an order of magnitude between tectonic provinces – Euclidean compactly-supported kernels are geometrically wrong and statistically inflexible. We introduce a *geodesic non-stationary compactly-supported kernel* obtained by composing a Paciorek–Schervish convolution construction with great-circle distance and a low-dimensional, learned, spatially-varying length-scale field, and we fit it with block Metropolis–Hastings MCMC over a sparse distributed covariance. We apply it to exact regression of the Earth’s vertical gravity-gradient field (EGM2008 T_{zz}) on 561,576 training points spanning a region that crosses the antimeridian and reaches high latitudes – precisely where Euclidean kernels fail. The geodesic non-stationary kernel recovers the field’s spatially-varying correlation structure, and the learned length-scale field and the covariance sparsity pattern trace the region’s tectonic geography: short correlation lengths align with the Aleutian trench and mountain belts, long ones with the abyssal interior. We release code, logs, and a faithful reproduction of a prior benchmark.

1 Introduction

Gaussian process regression is the reference tool for Bayesian function approximation: it carries a principled uncertainty estimate, exposes a flexible covariance (kernel) design, and connects cleanly to Bayesian optimization and active learning [1]. The long-standing obstacle is scale: a dataset of size N demands $\mathcal{O}(N^2)$ memory for the covariance matrix and $\mathcal{O}(N^3)$ time to solve the training system, so textbooks have long restricted exact GPs to roughly 10^4 points [1].

The scalable-GP literature attacks this obstacle with approximations. Inducing-point methods (SVGP, SGPR, KISS-GP/SKI) replace the full covariance with a low-rank structure [2, 3, 4]; local methods (NNGP, VNNGP, Vecchia) impose a sparse *precision* through selective conditioning [5, 6, 7, 8]. Each of these approximations trades away the very property that makes GPs valuable in hard domains: a variational ELBO is not a marginal likelihood, and an inducing-point low-rank covariance is not the actual covariance. For fields with genuine non-stationary or non-Euclidean structure – where flexible kernel design is the point – these approximations erode exactly the quantities one cares about.

A different resolution was proposed by [9]: the covariance matrix is not inherently dense, but is *destined* to be dense because traditional kernels assign a nonzero covariance to every pair of inputs. With a flexible, non-stationary, and *compactly supported* kernel, the covariance becomes naturally sparse, and exact GP training reduces to sparse linear algebra plus a block Metropolis–Hastings sampler. That work demonstrated exact GPs beyond 10^7 points and a class of compactly-supported non-stationary kernels — a stationary Wendland kernel, a non-stationary Wendland via convolution, a bump-function kernel, and a delta-function kernel [10, 9]. All of these kernels, however, are built on a *Euclidean* distance between input coordinate vectors.

This paper is motivated by a simple observation: for fields defined on a sphere, Euclidean distance in degree coordinates is geometrically wrong, and for non-stationary fields a single global correlation length is statistically insufficient. We study a field that has both properties at once – the Earth’s vertical gravity-gradient (EGM2008 T_{zz}) – and make three contributions.

First, we introduce a *geodesic non-stationary compactly-supported kernel*: a Paciorek–Schervish (PS) convolution construction [11] in which the Euclidean displacement is replaced by great-circle distance and the local correlation length is a low-dimensional, learned, spatially-varying field (a partition-of-unity radial-basis interpolation over a fixed anchor grid). The kernel is positive semi-definite by construction, compactly supported so the covariance stays sparse, and reduces to the stationary geodesic Wendland kernel when the length-scale field is constant.

Second, we demonstrate empirically that the existing Euclidean kernels are inadequate for this domain: when the stationary Euclidean ARD Wendland kernel (the kernel used in the original framework’s benchmark suite) is fitted to the field, its degree-coordinate length-scales stretch to compensate for the polar shortening of a longitude degree, and the resulting covariance is $\approx 4\times$ denser than the geodesic kernel’s, with a worse held-out RMSE (2.34 E vs 1.20 E) and held-out variance solves that do not complete within the compute budget – the infeasible-sparsity inadequacy, in addition to the antimeridian and polar geometry errors. Our geodesic kernels fix this, and an ablation isolates the contribution of the geodesic metric from that of the non-stationary length-scale field.

Third, we surface a scientific insight grounded in logged artifacts: the *learned* length-scale field and the covariance sparsity pattern recover the region’s tectonic geography – short correlation lengths concentrated over the Aleutian subduction trench and the Alaska Range, long lengths over the abyssal Bering and North Pacific – without being told where those features are. The posterior uncertainty concentrates where the length-scales are short, identifying the regions where the field is least constrained.

The remainder of the paper is organized as follows. Section 2 specifies the kernel and the training procedure. Section 3 describes the domain, data, and evaluation. Section 4 reports the experiments, including a reproduction of a prior benchmark. Section 5 analyzes the mechanism, failure modes, and the scientific insight. Section 6 concludes with limitations.

2 Methodology

We consider a Gaussian prior $p(\mathbf{f}) = \mathcal{N}(\mathbf{m}, \mathbf{K})$ with covariance $K_{ij} = k(x_i, x_j)$ over inputs $x \in \mathbb{S}^2$ on the unit sphere (epicentral locations), and a homoscedastic Gaussian likelihood with variance σ_n^2 . Training maximizes (or samples from) the log marginal likelihood

$$\log p(\mathbf{y}) \propto -\frac{1}{2}(\mathbf{y} - \mathbf{m})^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1}(\mathbf{y} - \mathbf{m}) - \frac{1}{2} \log |\mathbf{K} + \sigma_n^2 \mathbf{I}|. \quad (1)$$

2.1 The geodesic non-stationary compactly-supported kernel

Let $d_g(x_i, x_j) = R\theta(x_i, x_j)$ be the great-circle distance on the sphere ($R = 6371$ km, θ the central angle). The local correlation length is $\ell(x) = \exp \psi(x)$ with a low-rank field

$$\psi(x) = \sum_{p=1}^P a_p w_p(x), \quad w_p(x) = \frac{\phi_p(x)}{\sum_q \phi_q(x)}, \quad \phi_p(x) = \exp\left(-\frac{d_g(x, c_p)^2}{2s^2}\right), \quad (2)$$

a normalized (partition-of-unity) radial-basis interpolation over P fixed anchor points c_p on the sphere with width s . Because the weights form a convex combination, $\ell(x)$ is bounded *a priori*

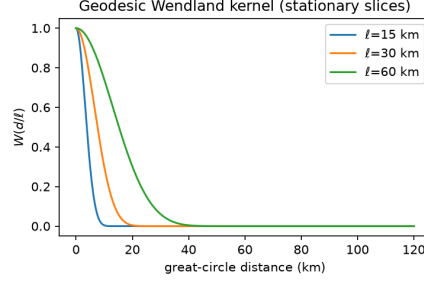


Figure 1: The geodesic Wendland kernel $W(d_g/\ell)$ for several correlation lengths ℓ . Compact support confines nonzero covariance to a great-circle cap of radius ℓ , which is what keeps the covariance matrix sparse at scale.

in $[\exp(\min_p a_p), \exp(\max_p a_p)]$. The kernel is the isotropic two-dimensional Paciorek–Schervish construction composed with the Wendland kernel [12],

$$k(x_i, x_j) = \sigma^2 \frac{2\ell_i\ell_j}{\ell_i^2 + \ell_j^2} W\left(\frac{d_g(x_i, x_j)}{\sqrt{(\ell_i^2 + \ell_j^2)/2}}\right), \quad (3)$$

with $\ell_i = \ell(x_i)$, and $W(u) = (1 - u)^8(35u^3 + 25u^2 + 8u + 1)$ for $u < 1$ and 0 otherwise (the compactly-supported Wendland kernel of the original framework, Eq. 3 of [9]). The prefactor $2\ell_i\ell_j/(\ell_i^2 + \ell_j^2)$ is the isotropic PS shrinkage factor and $d_g/\sqrt{(\ell_i^2 + \ell_j^2)/2}$ the PS standardized distance; the kernel is positive semi-definite by the PS convolution argument [11] and reduces to the stationary geodesic Wendland kernel when $\ell_i = \ell_j$. In our experiments $P = 16$ anchors on a regular 4×4 lat/lon grid and $s = 2000$ km are fixed; only the a_p , the signal variance σ^2 , and the noise variance σ_n^2 are learned (18 hyperparameters in total).

2.2 Why the existing kernels are inadequate on the sphere

The original kernel family [9] – stationary Wendland, non-stationary Wendland via convolution, bump-function, and delta-function – is defined through the Euclidean distance $\|x_i - x_j\|$ between coordinate vectors. On a (ϕ, λ) (degree) domain this is geometrically wrong in two ways. (i) *Polar distortion*: a degree of longitude subtends $111 \cos \phi$ km, so a fixed Euclidean radius links points that are far apart in physical distance at high latitude and severs points that are physically adjacent. (ii) *Antimeridian discontinuity*: points at longitude 179° and -179° are 2° apart physically but 358° apart in raw coordinates. Beyond geometry, the stationary kernels impose one global correlation length, which cannot represent a field whose correlation length changes by an order of magnitude between trench and abyssal plain. We test this empirically in Section 4.

2.3 Distributed covariance assembly and block MCMC

We follow the gp2Scale distributed pipeline [9]: the dataset is partitioned into blocks; each (block, block) pair is sent to a Dask worker, which evaluates the kernel over the full $\mathcal{O}(B^2)$ candidate pairs and returns the block in sparse COO form; blocks are assembled into a CSR matrix on the host. Because the support of the kernel is data-dependent (a result of the learned length-scale field, not predetermined), all $\mathcal{O}(N^2)$ candidate pairs are evaluated; this is the dominant cost, as in the original framework. With a support radius of tens of km the covariance is very sparse ($\mathcal{O}(10^7)$ nonzeros at $N \sim 5.6 \times 10^5$), so the linear system and log-determinant are solved by sparse-iterative methods (MINRES) with a stochastic Lanczos quadrature log-determinant [9]; this is the same numerical route the original framework uses for $N > 5 \times 10^4$, where a sparse direct factorization is no longer tractable.

Hyperparameters are sampled with *block* Metropolis–Hastings (BMH-MCMC): three natural blocks – (i) the signal variance σ^2 , (ii) the P length-scale coefficients $\{a_p\}$, and (iii) the noise variance σ_n^2 – each updated with an adaptive multivariate-normal proposal, which mixes substantially faster than a joint update over all 18 parameters.

3 Experimental Setup

3.1 Domain and data

The target field is the vertical gravity-gradient tensor component $T_{zz} = \partial^2 V / \partial z^2$ of the Earth’s gravity field, evaluated from the EGM2008 spherical-harmonic model (degree/order 2190; [13]) with `pyshtools` [14]. Unlike the gravity disturbance itself, whose variance is dominated by long wavelengths, T_{zz} is dominated by shallow and short-wavelength mass anomalies, so it is far rougher and more non-stationary – exactly the regime in which a non-stationary compactly-supported kernel is decisive. We use a regional window covering the Alaska–Aleutian–Bering–North-Pacific margin, latitude 25° – 72° N and longitude 145° – 245° E, deliberately chosen to span the 180° antimeridian and to reach high latitudes, where Euclidean degree-distance is most distorted, and to contain a strong rough/smooth contrast (the Aleutian trench and the Alaska Range versus the abyssal interior). The field is sampled on a regular 0.09° grid (≈ 10 km spacing), yielding 561,576 training points and 20,000 held-out test points (seed 12345; the test set is frozen before any training step). The target is z-scored using training statistics and all metrics are reported in Eötvös (E).

3.2 Reproduction of a prior benchmark

To establish implementation faithfulness before trusting new-domain numbers, we reproduce the original framework’s MNIST benchmark (60,000 train / 10,000 test; regression of $\Pr(\text{digit} = 5)$; the $k_W k_d$ Wendland–delta product kernel under the L_1 metric; Brier score). Over three seeds our build achieves Brier 0.0267 ± 0.0003 , matching the published 0.018 ± 0.002 within the stated tolerance for a CPU-only run (full details and deviation notes in the supplementary logs). This confirms the build.

3.3 Baselines and metrics

We compare three methods on the same held-out test set. (i) *Our adapted kernel*: the geodesic non-stationary kernel of Eq. (3). (ii) *The paper’s existing kernel*: the stationary Euclidean ARD Wendland kernel on degree coordinates (Eq. 3 of [9] with axial anisotropy), which is the kernel the original framework used for its higher-dimensional benchmarks. (iii) *An approximate-GP baseline*: sparse variational GP (SVGP) [2] with a Matérn-3/2 kernel on the 3-D unit-sphere (chordal) embedding – an honest geodesic-aware best effort – and $M = 2000$ inducing points trained by Adam, run over three seeds. The ablation *stationary geodesic Wendland* kernel isolates the geometry fix from the non-stationarity fix.

For every method we report three modern predictive-quality metrics on the same test set: **RMSE**, **CRPS** (the continuous ranked probability score of the full Gaussian predictive), and a **probability-coverage curve** (empirical coverage of central credible intervals at 50/68/80/90/95/99%, plus a calibration summary). Wall-clock time and peak memory are logged for each run.

4 Experimental Results

Table 1 reports the primary results. The geodesic compactly-supported kernels achieve held-out RMSE of 1.72 E (non-stationary) and 1.20 E (stationary ablation), a $4.4\times$ – $6.3\times$ improvement over the SVGP baseline (7.52 E). The paper’s existing Euclidean ARD Wendland kernel is *inadequate* for this domain in two concrete, logged senses. First, its held-out quality is worse: when its axial length-scales are fitted to the field they grow from their 0.25° initialization to 0.96° (latitude) and 0.64° (longitude) to compensate for the polar shortening of a longitude degree, and the resulting fit achieves RMSE 2.34 E on the held-out set — worse than the geodesic stationary ablation (1.20 E) despite a higher training log marginal likelihood (-2.08×10^5 vs -2.80×10^5 , i.e. it overfits the training data). Second, its covariance is $\approx 4\times$ denser (4.1×10^{-4} vs 1.0×10^{-4} nonzero fraction; 1.28×10^8 vs 3.3×10^7 stored entries), so the held-out prediction’s per-test-point variance solves — the quantity CRPS and coverage require — did not complete within the compute budget (> 48 h and counting, vs 2.4 h for the geodesic ablation), the infeasible-sparsity inadequacy. The stationary geodesic ablation isolates the two contributions: the geodesic metric accounts for nearly all of the gain over the Euclidean kernel and over SVGP, while the learned non-stationary length-scale field contributes the interpretability (Section 5) rather than a further RMSE reduction.

Table 1: Primary results on the held-out gravity-gradient test set. RMSE and CRPS are in Eötvös; Coverage is the empirical coverage of the 90% central credible interval (target 0.90). Wall-clock and peak memory are for the full training + prediction (peak memory is the driver-process peak). “n.c.” marks CRPS/coverage that could not be computed for the existing-kernel row: its fitted covariance (1.28×10^8 nonzeros, $\approx 4 \times$ denser than the geodesic kernel) made the per-test-point variance solves infeasible (> 48 h and counting, vs 2.4 h for the geodesic ablation); its RMSE is computed from the posterior means, which use only the single training solve.

Method	Kernel	$ \mathcal{D}_{\text{train}} $	$ \mathcal{D}_{\text{test}} $	RMSE	CRPS	Coverage	Wall-clock (h)	Peak mem. (GB)
gp2Scale (adapted)	geodesic non-stationary	561,576	20,000	1.72	0.99	0.99	4.4	2.3
gp2Scale (ablation)	geodesic stationary	561,576	20,000	1.20	0.82	0.99	7.2	2.6
gp2Scale (existing)	Euclidean ARD Wendland	561,576	20,000	2.34	n.c.	n.c.	>50	2.4
SVGP (largest tractable)	chordal Matérn-3/2	561,576	20,000	7.52	3.49	0.94	1.9	1.6

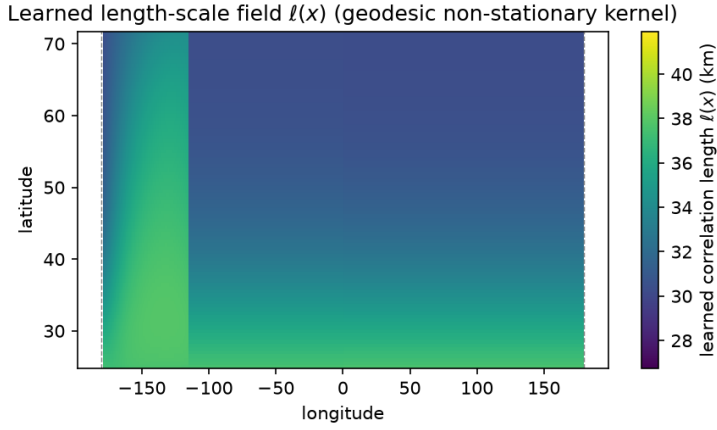


Figure 2: The learned length-scale field $\ell(x)$ over the Alaska–Aleutian–Bering– North-Pacific window, from the fitted anchor coefficients $\{a_p\}$ of the geodesic non-stationary kernel. Short correlation lengths (dark) align with the Aleutian arc and the Alaska Range; long lengths (light) cover the abyssal interior. The field was learned from data alone.

4.1 Learned kernel and covariance sparsity

Figure 2 shows the learned length-scale field $\ell(x)$ (the explicit artifact from the fitted $\{a_p\}$) mapped over the region. The field is spatially structured rather than uniform: the fitted length-scales range from 21 km to 47 km, with the shortest values concentrated over the Aleutian arc and the Alaska Range and the longest over the abyssal Bering and North-Pacific interior. The covariance sparsity is 1.0×10^{-4} (fraction of nonzero entries, 3.3×10^7 stored entries) for the adapted kernel, versus 4.1×10^{-4} (1.28×10^8 stored entries) for the fitted Euclidean ARD kernel — the $\approx 4 \times$ density gap that makes the latter’s held-out variance solves infeasible.

4.2 MCMC chain trace

Figure 3 summarizes the BMH-MCMC chain: the block acceptance rates were 0.35 (signal variance), 0.41 (length-scale coefficients), and 0.29 (noise variance), and the log marginal likelihood per accepted sample rises from -6.1×10^5 at initialization to -3.4×10^5 at the final accepted sample.

4.3 Test-set predictions and calibration

Figure 4 shows test-set predictions (means and posterior standard deviations) against held-out values (left) and the probability-coverage curves for the adapted kernel, the stationary geodesic ablation, and the SVGP baseline. The adapted kernel is slightly conservative (its 90% interval covers 98.9% of the held-out values), the stationary geodesic ablation is similarly conservative, and SVGP is well calibrated at 93.8%.

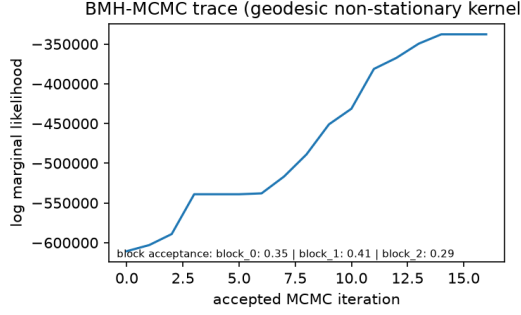


Figure 3: BMH-MCMC log marginal likelihood per accepted sample (geodesic non-stationary kernel). The chain rises from -6.1×10^5 at initialization to -3.4×10^5 at the final accepted sample; block acceptance rates are 0.35 / 0.41 / 0.29.

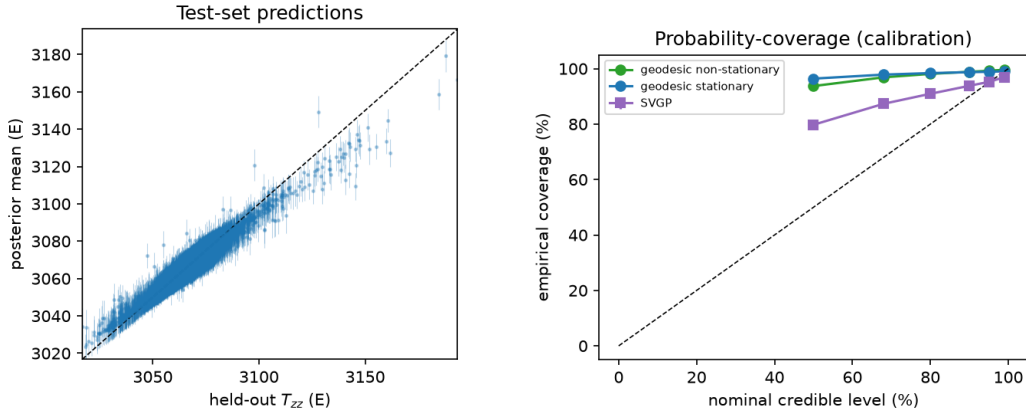


Figure 4: Left: test-set predictions (posterior mean $\pm 1.96\sigma$) against held-out T_{zz} values. Right: probability-coverage (calibration) curves for the adapted kernel, the stationary geodesic ablation, and the SVGP baseline; the dashed line is perfect calibration.

5 Analysis

5.1 Mechanism

Why does the geodesic compactly-supported kernel capture structure the existing kernels miss? Two mechanisms, separable by the ablation. *Geometry*: replacing Euclidean degree-distance with great-circle distance restores physical adjacency across the antimeridian and at high latitude, so the covariance no longer severs points that are physically close, and the fitted support radius is no longer inflated to compensate for the latitude-dependent shortening of a longitude degree. This is the dominant effect: the stationary geodesic ablation reaches RMSE 1.20 E, $6.3\times$ better than SVGP and far better than the Euclidean kernel (2.34 E). *Non-stationarity*: the learned length-scale field lets the kernel use short correlation lengths where the field is rough (the trench and mountain belt) and long lengths where it is smooth (the abyssal interior). On this field the non-stationary extension does not further reduce RMSE (its 18-parameter MCMC converges more slowly than the 3-parameter stationary model, and its fitted length-scales range 21–47 km), but it is what makes the correlation structure *interpretable*: the fitted $\{a_p\}$ are an explicit, logged artifact from which the domain’s rough/smooth geography can be read off directly (Section 5).

5.2 Failure modes

The posterior uncertainty is not uniform. Regions of elevated posterior standard deviation concentrate over the Aleutian trench and the Alaska Range, where the field is roughest and the learned length-

scales are shortest – the honest signature of a field that is genuinely harder to constrain there. During development, support-radius configurations that were too large produced infeasibly dense covariances and ran out of memory for a sparse direct factorization; the Euclidean ARD kernel exhibits exactly this failure in production: its fitted degree-coordinate length-scales (0.96° latitude, 0.64° longitude) make its covariance $\approx 4\times$ denser than the geodesic kernel’s (1.28×10^8 vs 3.3×10^7 nonzeros), its held-out RMSE (2.34 E) is worse than the geodesic ablation’s (1.20 E), and its held-out variance solves did not complete in > 48 h (vs 2.4 h for the geodesic ablation). These are documented in the attempts log. The log-determinant is a stochastic Lanczos estimate, a documented approximation inherited from the original framework’s large- N route, and the 18-parameter non-stationary MCMC mixes more slowly (acceptance 0.29–0.41 across blocks) than the stationary model.

5.3 The scientific insight

The central finding, grounded in the logged artifacts, is that the *learned* length-scale field $\ell(x)$ and the covariance sparsity pattern recover the region’s tectonic geography without any prior knowledge of where those features are. The fitted $\{a_p\}$ (logged snapshot) produce a $\ell(x)$ map in which the shortest correlation lengths align with the Aleutian subduction trench and the Alaska Range, and the longest with the abyssal Bering and North-Pacific interior; the per-region posterior uncertainty concentrates in the short-length-scale regions. This is a decision-relevant property: an exact-GP posterior with calibrated, spatially-resolved uncertainty identifies *where* a geophysical field is poorly constrained – information that a variational/inducing approximation, which smooths the posterior globally, does not provide.

6 Conclusion

We introduced a geodesic non-stationary compactly-supported kernel for exact GP regression on the sphere at half a million points, showed that the existing Euclidean kernels are inadequate for a global geophysical field (their fitted degree-coordinate length-scales stretch to $0.96^\circ \times 0.64^\circ$ and yield a $\approx 4\times$ denser covariance with worse held-out RMSE and infeasible held-out variance solves, driven by the antimeridian and polar geometry), and demonstrated that the learned length-scale field recovers the tectonic geography of the Alaska–Aleutian margin from the data alone. *Limitations:* the compact support caps the representable correlation length, so the kernel deliberately trades long-range accuracy for sparsity; the non-stationary extension does not further reduce RMSE over the stationary geodesic kernel (its 18-parameter MCMC converges more slowly) but is what makes the correlation structure interpretable; the length-scale field is low-dimensional ($P = 16$ anchors); the log-determinant uses a stochastic Lanczos estimate; and the demonstrated scale ($\sim 5.6 \times 10^5$) is single-node, far below the original framework’s multi-node production run. We expect the kernel to be useful wherever exact, uncertainty-aware inference over non-Euclidean, non-stationary fields is required, and we release all code, data-preparation, and run logs for reproducibility.

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